SANT LONGOWAL INSTITUTE OF ENGINEERING AND TECHNOLOGY
(Deemed-to-be University under MoE, Govt of India)
Longowal, Punjab – 148106
techFEST'26
SLIET Longowal's Annual Technical Fest

PATRON-IN-CHIEF
Prof. Mani Kant Paswan
Director, SLIET

PATRON
Prof. Vijay Kumar Kukreja
Dean (SW), SLIET

CHAIRMAN
Dr. Sunil Kumar
Associate Professor (ME)

VICE-CHAIRMAN
Dr. Barasha Mali
AP, (EIE)

VICE-CHAIRMAN
Dr. Tajinder Singh
AP, (CSE)

Date : **Ref. No.:**

THE AQUA-EPOCH
PROBLEM STATEMENT AND EVENT GUIDELINES

1. Introduction :

Welcome to The Aqua-Epoch! Global water quality is facing critical challenges due to pollution and climate change. Modern sensor networks collect large amounts of chemical and physical water data, but much of it remains uninterpreted. In this challenge, your goal is to analyze water quality data, build smart machine learning models to determine whether water is safe to drink, and develop a user-friendly web application for real-time predictions.

2. Event Details :

- Event Domain: Environmental Intelligence / Machine Learning & Web Development.
- Tech Stack Options: Python (Pandas, Scikit-Learn, XGBoost), React / Streamlit, Flask / FastAPI, PostgreSQL / Firestore.

3. Eligibility :

- Anyone who is a student enrolled in Diploma or Undergraduate program can participate.
- Participation is open to eligible students irrespective of their branch/discipline, subject to the event registration requirements.
- A team of 2 or 3 members candidates can make participation.
- Solo participations are allowed and upto 4 members are allowed for UG and Diploma 1st year only.
- All participants must register for the event through Unstop.
- Participants must provide accurate and valid student details during registration and submission.

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4. Predictive Tasks :

Task A — Classification (Water Safety / Potability)
Build a model (such as Random Forest or XGBoost) to classify water as Safe (1) or Unsafe (0). You must handle missing values and target class imbalances effectively.

Task B — Regression (pH Estimation)
Predict the continuous pH value of a water sample using the remaining chemical features, accounting for complex chemical interactions.

5. System & Web Application Requirements :

1. Frontend: A simple input form for water metrics (Turbidity, Hardness, TDS) with helpful tooltips on safe limits.
2. Input Validation: Ensure entered values fall within realistic physical ranges before submitting data to the backend.
3. Backend API: A REST API that processes user inputs and returns real-time model predictions.
4. Database (Bonus): Optionally log past predictions in a database for trend analysis and tracking.

• Dataset Link: Will be uploaded soon.

6. Evaluation Metrics :

Task / Track	Evaluation Metrics	Key Focus Area
Task A: Classification	F1-Score, ROC-AUC	Handling missing data, reducing false positives
Task B: Regression	RMSE, MAE	Accurate continuous pH prediction
System Implementation	Latency & UI Quality	Smooth API response, user experience, input validation

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7. Rules :

- Participants must register on the official techFEST'26 website.
- Models must be built from scratch using core machine learning libraries (e.g., Scikit-Learn, XGBoost) and should not rely on pre-built Generative AI or LLM APIs for core predictions.
- Shortlisted/qualified teams must bring their own laptops to demonstrate their model and working prototype during Round 2.
- Any violation of rules or misconduct will lead to immediate disqualification.
- The decisions of the organizers will be final and binding.
- Disqualified contestants will not receive a participation certificate.

8. Certification :

- Top 3 performers will receive winner certificates and cash prizes.
- All contestants (except those disqualified) will receive participation certificates.

9. Submission Guidelines :

- Round 1 (Technical Deck PDF): Submit a presentation covering Data Preprocessing, EDA, Machine Learning Models, and System Flow Diagram.
- Round 2 (Live Working Prototype): Demonstrate your deployed web application with real-time predictions and visualizations.

Important Note :

Rules and guidelines are subject to change, and participants are advised to refer to the official techFEST'26 website for updates and instructions. Any alterations will be communicated through the website, and adherence to the latest guidelines is mandatory for all participants.

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for any further query reach out to:

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Join Us!

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